

T3 Architectural Concept Disambiguation Synthesis

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This work derives from and formalizes commitments across the CKS theory trilogy: *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026), *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026), and *Inter-Self Coordination via Shared Substrate / Full Aspect Integration* (Li, April 2026).

Abstract

This note is the T3 subseries synthesis. Notes T3.01 through T3.04 each addressed one architectural concept — Self, aspect/cell, layer, and configuration — and established how that concept is used across the three CKS theory papers. This synthesis consolidates the four disambiguations into a single finding: the trilogy's architectural concepts are stable. They carry the same definitions in all three papers. Paper 3 does not redefine the architecture it inherits; it extends the roles some concepts play at larger coordination scopes and introduces specific named artifacts that are instances of general architectural principles already established in earlier papers. This finding — architectural stability — is what makes the trilogy coherent as a single architecture rather than three independent papers.

1. The T3 Subseries: What Was Established

Notes T3.01 through T3.04 covered four architectural concepts that appear across multiple papers in the CKS trilogy. The question each note addressed was the same: does this concept mean the same thing in Papers 2 and 3 as it did when first introduced, or does the concept shift meaning as the architecture scales?

T3.01 — Self. Self is the three-tier organizational AI governance architecture — the unit comprising a governance hierarchy of cells, aspects, and a Self-level substrate, governed by human-authored rules at all three tiers. Paper 2 introduced this architecture. Paper 3 uses Self as the participation unit for Full Aspect Integration (FAI) events without redefining it. The only Paper 3 addition specific to Self is the minimum valid Self specification (D6.04), which states the conditions a Self must satisfy to participate in FAI. That addition specifies a threshold for participation; it does not alter what a Self is.

T3.02 — Aspect and Cell. The cell is the most stable concept in the trilogy. Its definition — the atomic coordination unit executing a specific informational task under orchestration rules within a governance perimeter — is unchanged from Paper 1 through Paper 3. Paper 2 articulated internal cell structure (DNA layer, action layer, lifecycle, content-domain specification) that was implicit in Paper 1; it did not change the cell's definition. Paper 3 has cells continuing to operate within each Self's home perimeter exactly as Papers 1 and 2 specify; cells participate in inter-Self coordination indirectly, through their parent aspects. The aspect, introduced in Paper 2 as the coordination layer grouping cells by content-domain within a Self, acquires an additional role at Paper 3 scope: it becomes the unit of exchange in FAI. When a Self contributes to the shared substrate, it contributes aspects. The role expands; the definition does not change.

T3.03 — Layer. The four layer concepts — instinct layer, DNA layer, action layer, and reasoning layer — are consistently defined across Papers 2 and 3. Paper 2 introduced the instinct/reasoning separation as the central architectural commitment and named the four layers. Paper 3 inherits those definitions without alteration and adds one cross-perimeter governance rule: reasoning-layer content may cross Self perimeters during FAI (it is the exchange currency); instinct-layer content may not. This addition is a governance rule at the inter-Self boundary, not a redefinition of what any layer is.

T3.04 — Configuration. Configuration names the general principle that governance content for a coordination unit is authored as explicit substrate content — rules, constraints, purpose statements, and scope specifications that govern how that unit behaves. This principle is consistent throughout the trilogy. Paper 3 introduces a specific named artifact, FAI configuration, which is the six-dimension governance document specifying the terms of a particular FAI event. FAI configuration is one instance of the general configuration principle applied at the inter-Self scope; it does not alter the general principle.

2. The T3 Pattern

Across all four disambiguations, the same three-element pattern emerges.

Definitions consistent. The architectural concepts have the same definitions in Papers 2 and 3 as they had when first introduced. Self means what Paper 2 established. Cell means what Paper 1 established. Aspect means what Paper 2 established. Layer concepts mean what Paper 2 established. Configuration as a principle means what was already operative in Papers 1 and 2. Paper 3 does not redefine any of these concepts.

Roles extended. Some concepts acquire additional roles when the architecture operates at larger coordination scopes. The aspect gains the role of inter-Self exchange unit at Paper 3 scope while retaining its within-Self coordination role from Paper 2. The layer concepts gain cross-perimeter governance rules specifying which content may cross Self perimeters. In both cases, the concept's architecture is unchanged; its operational reach expands.

Artifacts added. Paper 3 introduces specific named artifacts that are instances of general architectural principles already in place. FAI configuration is the clearest example: the general principle (authored governance content for a coordination unit) was established in Papers 1 and 2; Paper 3 names a specific artifact of that type for the FAI context. The artifact is novel; the principle it instantiates is not.

These three elements — consistent definitions, extended roles, added artifacts — are not the same kind of finding. Consistent definitions establish that the prior art from Papers 1 and 2 covers Paper 3's usage of the same concepts. Extended roles establish that the concept's larger-scope applications inherit from, rather than depart from, the concept's established architecture. Added artifacts establish that Paper 3's named constructs are instances of general principles that were already public prior art before Paper 3 was written.

3. The T3 Consistency Claim

The T3 subseries establishes a single architectural claim about the trilogy: the three CKS papers operate on the same architectural foundation. When Paper 3 uses the terms Self, aspect, cell, layer, or configuration, those terms mean what Papers 1 and 2 established. The architecture does not restart at Paper 3; it extends.

This consistency has direct prior-art consequences. An attempt to claim novelty by asserting that Paper 3's "Self" is a distinct architectural object from Paper 2's "Self" — or that Paper 3's use of aspect as an exchange unit is a new concept unrelated to Paper 2's aspect — is foreclosed by the four T3 disambiguation notes. Those notes formalize, concept by concept, that each architectural term carries a stable definition across the trilogy. Anyone implementing the architectural patterns Paper 3 describes is implementing Paper 1 and Paper 2 architecture at a larger scope.

The T3 consistency claim also has structural significance. The trilogy's coherence as a design pattern series depends on architectural stability. If each paper introduced new definitions for the same terms, the three papers would be three independent architectures that happened to share vocabulary. Because the definitions are stable and only the roles

and artifacts change, the three papers compose into one architecture at progressively larger coordination scopes — cell, Self, inter-Self, population.

4. Contrast with T1 and T2

The T3 finding of architectural stability is more complete than the analogous findings in T1 and T2. T1 addressed governance concepts — substrate, authority distribution, conflict handling, provenance, governance perimeter — and found scope-dependent variations: the same concept with meaningfully different applications at cell scope, Self scope, and inter-Self scope. The substrate at inter-Self scope is a shared substrate spanning multiple home perimeters; conflict handling at inter-Self scope adds an escalate tier to the preserve/resolve pair. These are the same concepts, but with scope-specific elaborations that a careful reader must track. T2 addressed coordination scope transitions along the substrate-mediator ladder and similarly found that each rung transition involves substantive scope expansion. T3 is different: the architectural concepts do not elaborate at each scope in the way governance concepts do. Self, cell, aspect, layer, and configuration mean the same thing regardless of which paper is being read.

5. Transition to T4

The T4 subseries covers operational concept disambiguation — how operational governance concepts used throughout the trilogy are consistent across papers with scope-specific applications. The operational concepts T4 addresses include record, account, provenance, inspection, the three rights (inspect, modify, override), and scope as a governance boundary specifier. These concepts are expected to display the same architectural stability that T3 established for architectural concepts. The T4 prior-art value is the operational analogue of T3's architectural claim: that Paper 3's use of "inspection" means what Papers 1 and 2 established, that "provenance" carries the same six-field accountability structure across all three papers, and that the three rights are the same authority commitment at every scope the trilogy covers.

T3 established that the trilogy's architecture is stable. T4 establishes that the trilogy's operational governance layer is equally stable. Together, T3 and T4 close the prior-art space against any claim that Paper 3 introduces novel concepts by reusing familiar terms in new ways.